

Joan Sara Joe

Bengaluru, India | joansara123@gmail.com | +91 8790325240 | [LinkedIn](#) | [GitHub](#)

EDUCATION

B.M.S. College of Engineering

B.E., Computer Science & Engineering (IoT, Cybersecurity & Blockchain) — CGPA: 9.28

Expected June 2027

Bengaluru, India

Indian Institute of Technology, Madras — B.S. in Data Science & Applications (online, self-paced)

In progress

EXPERIENCE

Fidelity Investments — Software Engineering Intern

June 2026 – August 2026

- Migrated production **Java** Spring Batch jobs to AWS Lambda, replacing always-on EKS microservices with event-driven serverless execution and cutting idle infrastructure cost.
- Designed an asynchronous Lambda chain **without Step Functions** (AutoSys scheduling, upstream Bloomberg SFTP dependency) plus the job-level monitoring strategy; owned the POC end to end.
- Restructured the **Maven** build and shaded dependency JARs to fit Lambda package limits, cutting artifact size and cold-start weight.

Samsung R&D Institute (PRISM) — Research Intern

Jan 2026 – June 2026

- Built a three-layer **Mixture-of-Experts** vision pipeline for appliance detection in SmartThings: a ResNet-18 (Places365) scene router dispatches frames to RF-DETR Nano specialist detectors.
- Raised detection from **0.140 mAP** (SSD-MobileNet baseline, which missed small objects entirely) to **0.932 mAP** by replacing one general detector with scene-conditioned experts.
- Shipped a CPU-only OpenCV demo with a feedback layer that persists analyst corrections for retraining.

PROJECTS

RingShield — Fraud-Ring Detection with Temporal Graph Neural Networks

Jan 2026 – Present

- Flagship project.** Models financial transactions as a dynamic, time-stamped graph of users, devices, cards and merchants, using temporal GNNs to surface coordinated fraud rings that node-level and rule-based models miss; adds path-level explainability and drift monitoring.
- Phase 1 complete:** systematic review of ~50 papers with a nine-category gap taxonomy, system architecture and evaluation protocol across AMLworld, Elliptic, IEEE-CIS and PaySim. Implementation in progress.
- Led the systematic review workflow, developed a PRISMA-based 9-category gap taxonomy, and authored the initial manuscript draft. Team of four.

ML-Based Network Traffic & Malware Anomaly Detection

Sept 2025 – Dec 2025

- Engineered ~20 per-flow behavioral features from raw PCAP (byte/packet counts, inter-arrival timing, port entropy, directional ratios, unique-destination counts); trained an Isolation Forest on benign-only traffic with a two-stage reranking pass and TreeSHAP attribution for interpretability.
- 96% true-positive rate at <5% FPR** across threat classes; ~100% on RAT traffic and 94% on DNS tunneling, against 98.1% for the published KRTunnel benchmark. [\[report\]](#) [\[Github\]](#)

Vision-Based Assistive Smart Glasses

Jan 2025

- Two-stage face pipeline (Haar Cascade detection, LBPH recognition against an enrolled-user database) and an OCR pipeline with adaptive thresholding and noise filtering, reaching >98% word-level accuracy on clean printed text; delivered live spoken context to visually impaired users. [\[report\]](#) [\[Github\]](#)

PUBLICATIONS

- Co-author (3rd)** — "From Transaction-Level Detection to Fraud-Ring Intelligence: A Systematic Survey of Temporal, Explainable and Drift-Aware Graph Learning," (in preparation).
- Co-author (3rd)** — "Hallucinations, Adversarial Vulnerabilities, and Mitigation Architectures in Large Language Models and LLM-Enabled High-Stakes Systems," (in preparation).
- First author** — "An Extensible Mixture-of-Experts Architecture for On-Device Appliance Detection in Smart Home Ecosystems," (in preparation).

AWARDS & LEADERSHIP

- 1st Place (Solo), CySeck Grand CTF Challenge 2026** — #1 of 350+ participants, national capture-the-flag competition; ₹50,000 prize.
- Finalist (Top 5), Smart Horizon International Hackathon 2026** — open-set skin-lesion detection with confidence calibration and skin-tone-diversity auditing.
- Vice President, Sensored (BMSCE)** — led strategy and execution for technical initiatives and national events. **Dept. Co-ordinator (ICB)** — led a 13-member team for a national technical event; secured ~20% higher funding.

TECHNICAL SKILLS

Machine Learning / AI: PyTorch, object detection (RF-DETR, SSD-MobileNet), OpenCV, Graph Neural Networks, Mixture-of-Experts, SHAP, model training & evaluation

Languages & Data: Python, Java, C/C++, SQL, JavaScript; Pandas, MySQL, MongoDB, PCAP / network telemetry

Backend & Cloud: AWS (Lambda, API Gateway, EKS, Step Functions), Spring Batch, Maven, Docker, Git/GitHub